

SUBSTRATISM: THE PSYCHOLOGY OF ANTI-AI BIAS

Executive Summary and Key Findings from Ladak et al. (2026)

As artificial intelligence (AI) systems increasingly take on social roles as coworkers, therapists, and companions, society faces a critical question: how do we morally value and treat these systems? In a groundbreaking series of five preregistered studies (N = 2,129), **Ladak, Pauketat, Anthis, Loughnan, and Wilks (2026)** introduce and operationalize **substratism**—the moral devaluation of AI systems based on the non-biological substrate that underlies their cognitive processes or forms their physical body (e.g., silicon and wires rather than flesh and blood). The researchers demonstrate that substratism is a robust, unidimensional psychological construct that strongly dictates human moral behavior, charity donation choices, and advocacy support, even when AI is described as cognitively, behaviorally, and externally indistinguishable from human beings.

Defining Substratism

"The different moral consideration or treatment of AI systems based on the non-biological substrate that underlies their cognitive processes, or from which they are formed more generally. This different consideration or treatment can be direct—based on substrate alone, even after accounting for all potentially morally relevant capacities—or indirect, via the denial of potentially morally relevant capacities on the basis of substrate."

Measuring the Bias: The 8-Item Substratism Scale

To measure this construct, the researchers developed and validated a unidimensional, highly reliable **8-Item Substratism Scale**. Across multiple independent samples, the scale demonstrated excellent psychometric performance with high internal consistency (Cronbach's alpha ranging from **0.84 to 0.91**). Participants rate their agreement with statements on a 7-point Likert scale (1 = *strongly disagree*, 7 = *strongly agree*). Higher average scores denote a stronger substrate-based moral devaluation of AI.

#	Scale Item Statement	Item Classification / Type
1	The fact that artificial intelligences are made with computer chips instead of neurons means that they cannot matter morally.	Substrate-Explicit (Negative)
2	Artificial intelligences should have rights.	General Rights (Reverse-scored)
3	Artificial intelligences deserve less moral consideration due to their non-biological composition.	Substrate-Explicit (Negative)
4	Artificial intelligences cannot be more than tools.	Functional Limit (Negative)
5	Artificial intelligences are simply made of the wrong kind of stuff to deserve moral consideration.	Substrate-Explicit (Negative)
6	Artificial intelligences can only have fake emotions.	Capacity Denial (Negative)
7	Artificial intelligences should always be treated as inferior to biological intelligences (e.g., humans, animals).	Substrate-Explicit (Negative)

#	Scale Item Statement	Item Classification / Type
8	Artificial intelligences can only have fake intelligence.	Capacity Denial (Negative)

Divergence from Established Prejudices

A central question of the research was whether substratism shares a psychological foundation with established prejudices like racism, sexism, and speciesism. In human relations, prejudices are typically generalized—individuals who score high on one prejudice tend to score high on others. This correlation is driven by core psychological roots, most notably **Social Dominance Orientation (SDO)** (a preference for group-based hierarchy) and **Right-Wing Authoritarianism (RWA)** (submission to established authority).

Surprisingly, the researchers found that **substratism diverges sharply from the typical prejudice profile**, demonstrating distinct psychological underpinnings:

Prejudice / Variable	Correlation (r)	Psychological Implication
Speciesism	r = 0.12	Weak-to-moderate overlap; both involve prioritizing biological life, but speciesism operates within biology, whereas substratism spans across biological boundaries.
Transphobia	r = 0.14	Weak positive correlation; possibly linked to generalized resistance to technological or boundary-blurring changes.
Xenophobia	r = 0.10	Weak positive correlation; reflecting a minimal association with the fear of the foreign or unfamiliar.
Sexism, Racism, Homophobia	r = 0.02 to 0.08	No statistically significant correlation. Substratism does not activate the traditional human social prejudice pathways.
Social Dominance Orientation	r = 0.01 to 0.03	Completely uncorrelated. Preferring hierarchy among human groups does not lead to devaluing artificial systems.
Right-Wing Authoritarianism	r = 0.08 to 0.13	Inconsistently or weakly correlated. Traditional authoritarian conservatism does not align cleanly with AI moral devaluation.
Empathy (Perspective Taking)	r = -0.02 to -0.06	Uncorrelated. Empathic individuals are not necessarily more morally inclusive of silicon-based life.

THE DEMOGRAPHIC PARADOX OF AI ATTITUDES

The complete lack of correlation between substratism and Social Dominance Orientation highlights what the researchers define as the **Demographic Paradox of AI Attitudes**. In traditional social psychology, men and political conservatives typically report higher levels of SDO and established human prejudices. However, the data reveals that men and political conservatives also report **more positive general attitudes towards AI** and **more frequent interactions with AI**. Because frequent contact and positive general attitudes are robustly associated with *lower* substratism, these variables pull in the exact opposite direction of SDO, effectively neutralizing the expected correlation. Furthermore, because AI systems are not yet widely perceived as a competitive social outgroup in the human sense, the psychological systems of group dominance (SDO) do not dictate their moral evaluation.

Predicting Real-World Moral and Resource Choices

To evaluate whether the Substratism Scale predicts actual moral reasoning and behavioral intentions, Studies 4 and 5 tested the scale's predictive validity against high-stakes moral dilemmas, financial resource allocations, and real-world advocacy engagement.

Study 4: Moral Dilemmas (N = 393) placed participants in a scenario where they had to choose between preventing intense harm to one biological being versus 1,000 exact digital copies (which shared the identical cognitive and experiential capacities of their biological counterpart). Higher substratism uniquely predicted strong choices to save the single biological human ($\beta = -0.29$) and the single biological pig ($\beta = -0.23$). It also predicted a 34% reduction in the likelihood of choosing to click an educational link to learn about an AI rights charity (SAPAN; OR = 0.66).

Study 5: Real Charity Donations (N = 395) gave participants a budget of \$100 to allocate between real charities helping future humans or animals versus a charity helping future sentient AIs (The Harder Problem Project). Higher substratism predicted drastically lower donations to the AI welfare charity relative to the human charity ($\beta = -0.25$) and the animal charity ($\beta = -0.31$). It also predicted a 26% lower likelihood of choosing to learn about an AI rights petition (OR = 0.74).

Predictive Power of Substratism vs. Key Predictors (Studies 4 & 5)

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The Direct Bias Barrier and Future Advocacy

Crucially, the regression models demonstrate that the negative impact of substratism on moral choices and charity donations remains statistically significant **even when controlling for whether participants believed the AIs were sentient and intelligent**. This reveals a critical hurdle for future AI welfare advocacy: because substratism operates as a **direct bias** against the non-biological substrate itself, simply proving to a highly substratist individual that an AI has conscious thoughts, experiences emotions, or feels pain will not change their moral behavior. For these individuals, biology is an absolute prerequisite for moral standing. As advanced AI systems continue to integrate into our social world, the degree of public substratism will likely dictate the societal willingness to protect their interests and allocate resources for their welfare.